

Class 14: RNASeq mini project

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Background

Here we work through a complete RNASeq analysis project. The input data comes from a knock-down experiment of a HOX gene

Data Import

Reading the `counts` and `metadata` CSV files

```
counts <- read.csv("GSE37704_featurecounts.csv", row.names = 1)
metadata <- read.csv("GSE37704_metadata.csv")
```

Check the structure

```
head(counts)
```

	length	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370
ENSG00000186092	918	0	0	0	0	0
ENSG00000279928	718	0	0	0	0	0
ENSG00000279457	1982	23	28	29	29	28
ENSG00000278566	939	0	0	0	0	0
ENSG00000273547	939	0	0	0	0	0
ENSG00000187634	3214	124	123	205	207	212
	SRR493371					
ENSG00000186092	0					
ENSG00000279928	0					
ENSG00000279457	46					
ENSG00000278566	0					
ENSG00000273547	0					
ENSG00000187634	258					

```
metadata
```

	id	condition
1	SRR493366	control_sirna
2	SRR493367	control_sirna
3	SRR493368	control_sirna
4	SRR493369	hoxa1_kd
5	SRR493370	hoxa1_kd
6	SRR493371	hoxa1_kd

Some book-keeping is required as there looks to be a mis-match between metadata rows and counts columns

```
ncol(counts)
```

```
[1] 7
```

```
nrow(metadata)
```

```
[1] 6
```

Looks like we need to get rid of the first “length:column of our counts objects.

```
cleancounts<-counts [,-1]
```

```
colnames(cleancounts)
```

```
[1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

```
metadata$id
```

```
[1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

```
all(colnames(cleancounts) == metadata$id)
```

```
[1] TRUE
```

Remove zero count genes

There are lots of genes with zero counts. We can remove these from further analysis.

```
head(cleancounts)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000186092	0	0	0	0	0	0
ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46
ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

```
to.keep.inds <- rowSums(cleancounts) > 0  
nonzero_counts <-cleancounts[to.keep.inds,]
```

##DESeq analysis Load the package

```
library(DESeq2)
```

Setup DESeq object

```
dds <-DESeqDataSetFromMatrix (countData=nonzero_counts,
                              colData= metadata,
                              design= ~condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

Run DESeq

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

Get results

```
res <- results (dds)
head(res)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 6 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.9136	0.1792571	0.3248215	0.551863	5.81042e-01
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630156	1.43993e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01

```

ENSG00000187642    11.9798      0.5428105 0.5215598    1.040744 2.97994e-01
                padj
                <numeric>
ENSG00000279457 6.86555e-01
ENSG00000187634 5.15718e-03
ENSG00000188976 1.76553e-35
ENSG00000187961 1.13413e-07
ENSG00000187583 9.19031e-01
ENSG00000187642 4.03379e-01

```

##Data Visualization

Volcano Plot

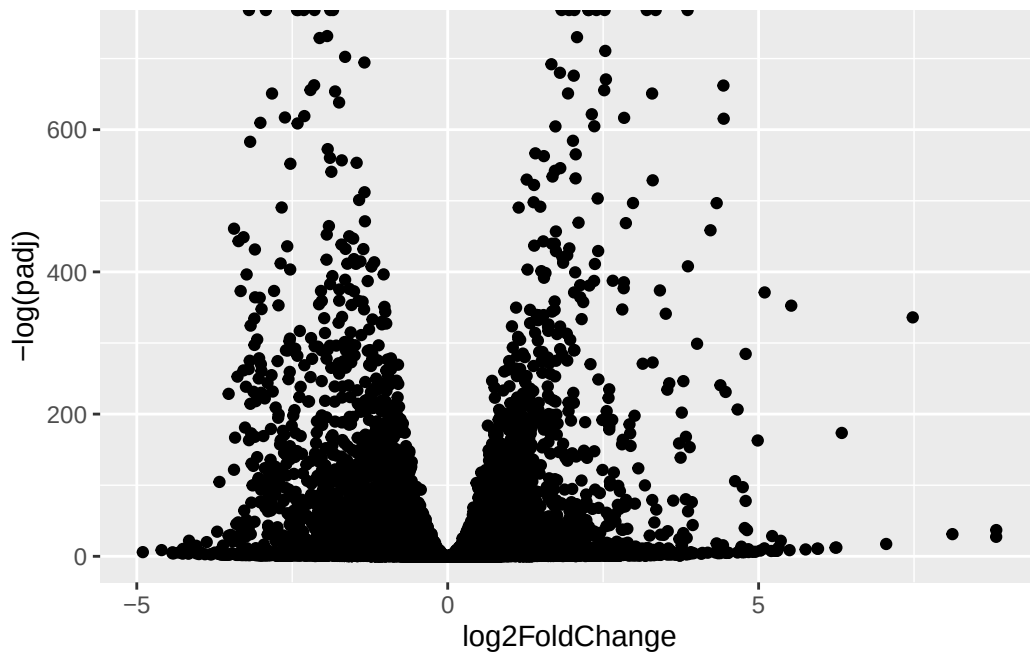
```

library(ggplot2)

ggplot(res) +
  aes(log2FoldChange, -log(padj)) +
  geom_point()

```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



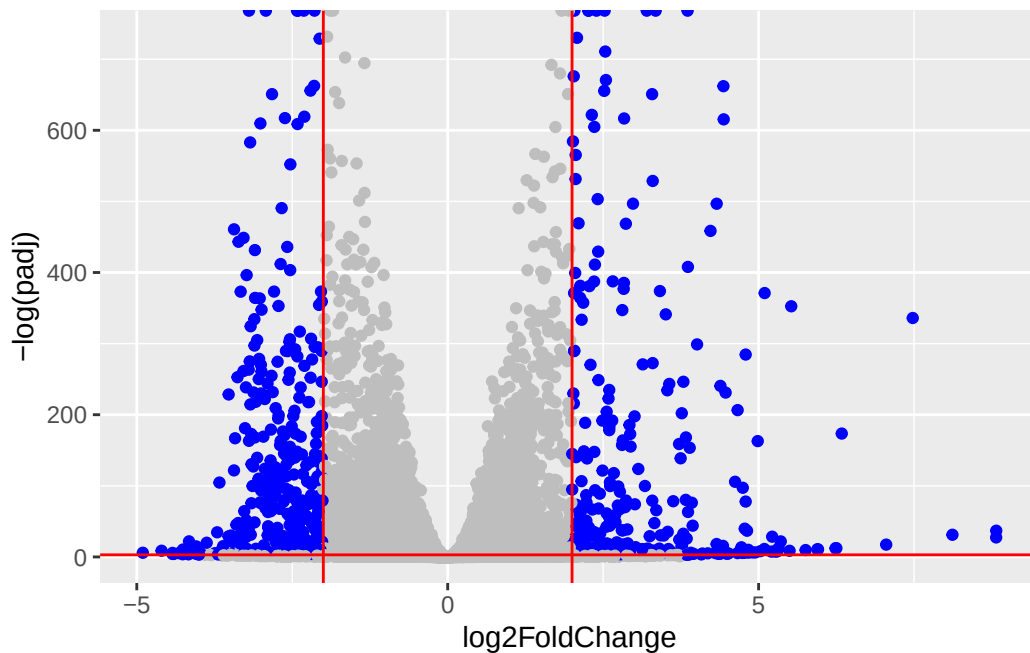
Add threshold lines for fold-change and p-value and color out subset of genes that make these threshold cut-pffs in the plot

```
library(ggplot2)

mycols <- rep("gray", nrow(res))
mycols[ abs(res$log2FoldChange) > 2 ] <- "blue"
mycols[ res$padj > 0.05 ] <- "gray"
mycols[ is.na(res$padj) ] <- "gray"

ggplot(res) +
  aes(log2FoldChange, -log(padj)) +
  geom_point(color = mycols) +
  geom_vline(xintercept = c(-2, 2), color = "red") +
  geom_hline(yintercept = -log(0.05), color = "red")
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



Add Annotation

Add gene symbols and entrez ids

```
library(AnnotationDbi)
library(org.Hs.eg.db)
```

```
res$symbol<- mapIds(x=org.Hs.eg.db,
                    keys= row.names(res),
                    keytype="ENSEMBL",
                    column="SYMBOL")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez<- mapIds(x=org.Hs.eg.db,
                    keys= row.names(res),
                    keytype="ENSEMBL",
                    column="ENTREZID")
```

'select()' returned 1:many mapping between keys and columns

Pathway Analysis

##KEGG Run Gage Analysis with KEGG

```
library(gage)
library(gageData)
library(pathview)
```

We need a named vector of fold-change values as input for gage

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

<NA>	148398	26155	339451	84069	84808
0.17925708	0.42645712	-0.69272046	0.72975561	0.04057653	0.54281049

```
data(kegg.sets.hs)
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

```
head(keggres$less, 2)
```

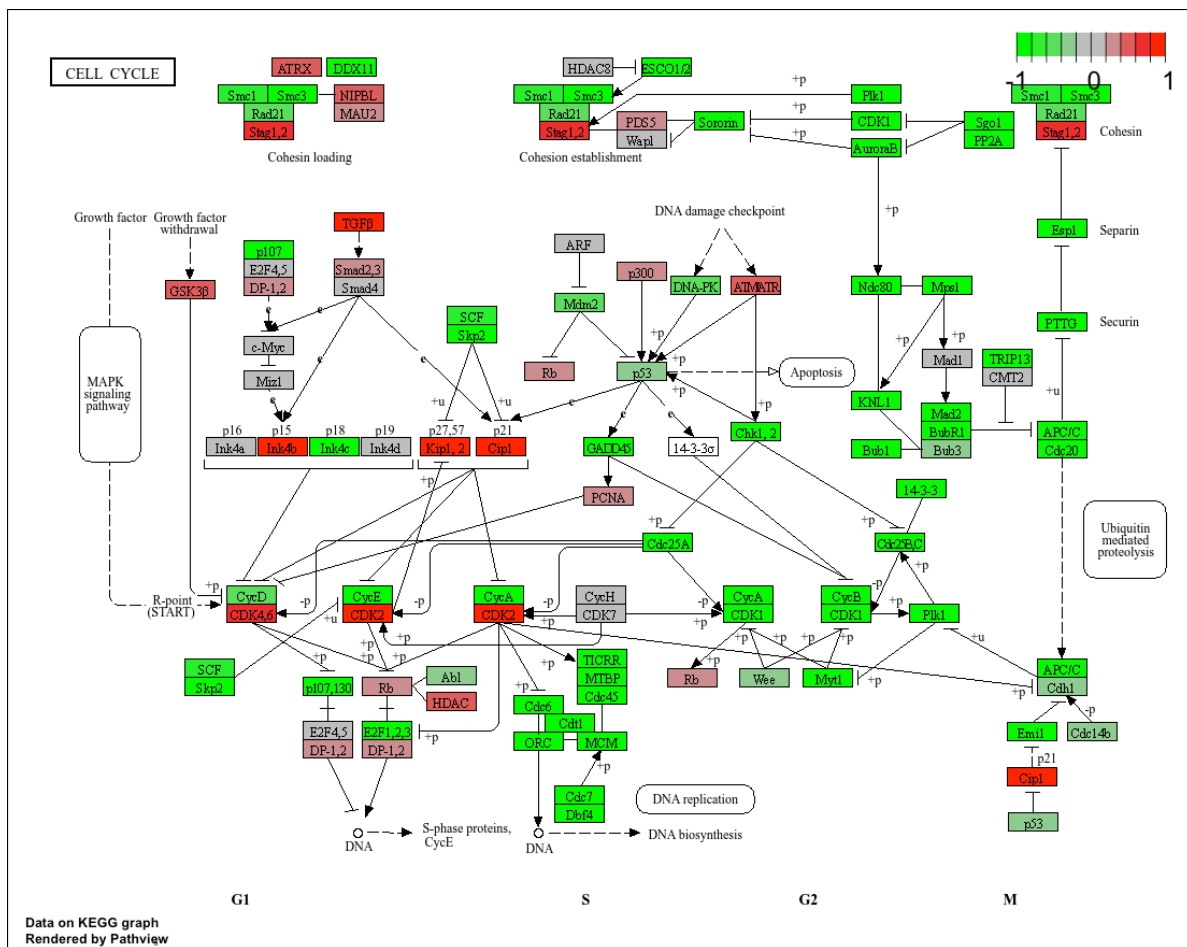
		p.geomean	stat.mean	p.val	q.val
hsa04110	Cell cycle	8.995727e-06	-4.378644	8.995727e-06	0.001889103
hsa03030	DNA replication	9.424076e-05	-3.951803	9.424076e-05	0.009841047
		set.size	exp1		
hsa04110	Cell cycle	121	8.995727e-06		
hsa03030	DNA replication	36	9.424076e-05		

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/chaewoochun/Desktop/BIMM143/class 14

Info: Writing image file hsa04110.pathview.png

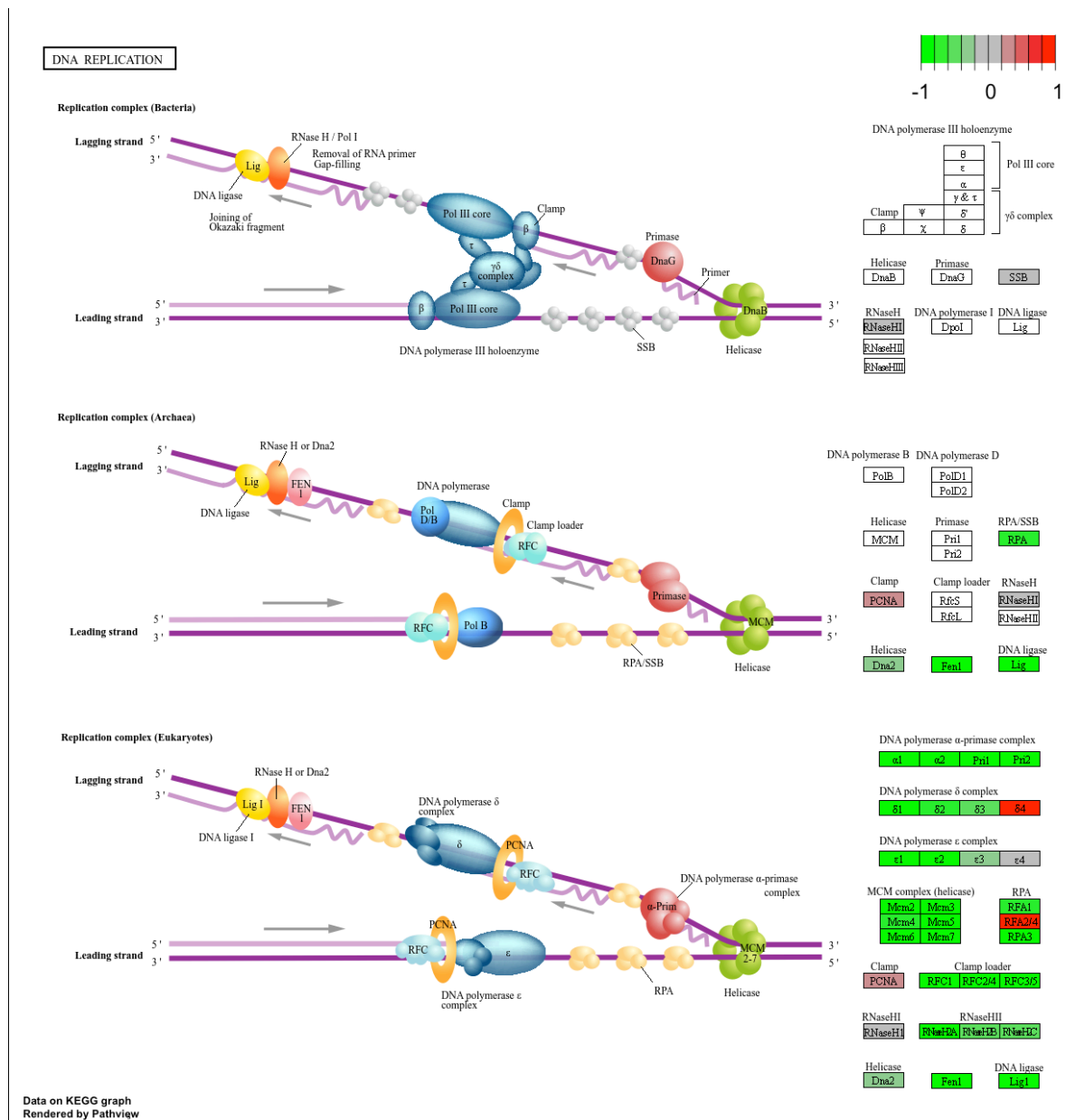


```
pathview(gene.data=foldchanges, pathway.id="hsa03030")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/chaewoochun/Desktop/BIMM143/class 14

Info: Writing image file hsa03030.pathview.png



GO terms

Same analysis but using GO genesetws rather than KEGG

```
data(go.sets.hs)
data(go.subs.hs)
```

```
# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)
```

```
head(gobpres$less, 4)
```

		p.geomean	stat.mean	p.val
G0:0048285	organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280	nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067	mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
		q.val	set.size	expl
G0:0048285	organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280	nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067	mitosis	5.841698e-12	352	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14

Reactome

Lots of folks like the reactome web interface. You can also run this as an R function but lets look at the website first < <https://reactome.org/user/guide>> the website wants a text file with one gene symbol per line for the genes you want to map to pathways.

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
head(sig_genes)
```

```
ENSG000000187634 ENSG000000188976 ENSG000000187961 ENSG000000188290 ENSG000000187608
      "SAMD11"          "NOC2L"          "KLHL17"          "HES4"          "ISG15"
ENSG000000188157
      "AGRN"
```

```
#res$symbol
```

and write out to a file:

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

Save our Results

```
write.csv(res,file="myresults.csv")
```